

Sheet G07 - Mesh Smoothing

Solutions for the theoretical and practical part via eCampus by Mo, 01.12.2025, **10:00**.

Practical Part

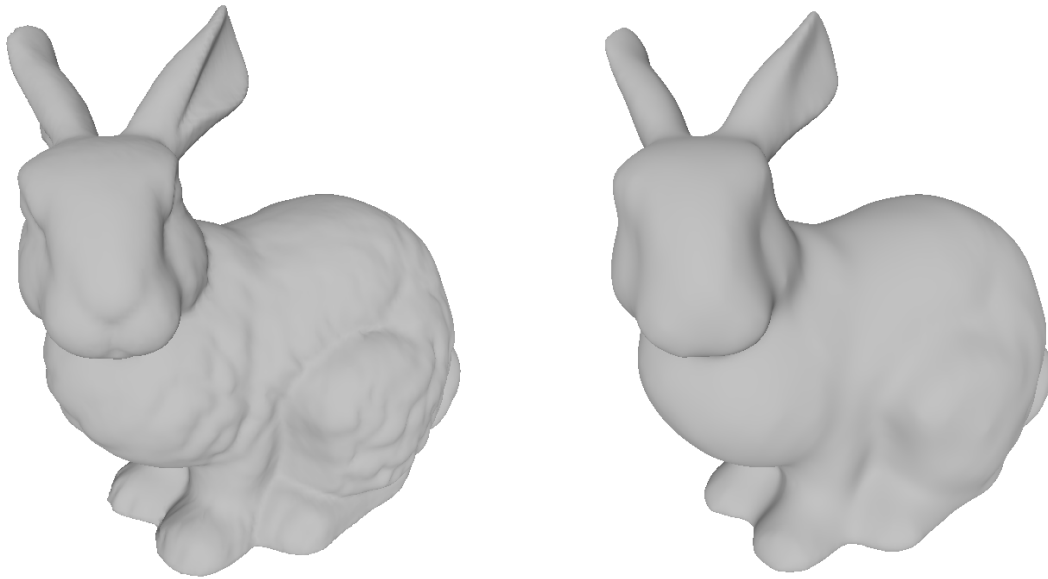


Figure 1: Smoothed bunny after 200 iterations.

Assignment 1) Taubin Smoothing

(6Pts)

Download and add `framework_g07.zip` to the project.

- In the struct `LaplaceUniform` implement the calculation of the Uniform Laplace ($w_{ij} = 1$). Make sure that the entries on the diagonal are negative.
- In the function `taubin_smoothing(...)`, retrieve the matrices S and M from your implementation and compute the Laplace matrix $L = M^{-1}S$. Use L to compute the Taubin Operator

$$X^{(k+1)} = (I + \mu L) \cdot (I + \lambda L) X^{(k)}$$

and implement one step of the iterative smoothing operation.

Submit this exercise by uploading the whole source file onto eCampus!

Theoretical Part

Upload the solution as pdf file on eCampus!

Assignment 2) Euler vs. Taubin Smoothing

(2Pts)

What is the main problem with simple Laplace smoothing via Euler iterations? How does Taubin Smoothing solve this problem? State the iteration formula of explicit Euler Smoothing and Taubin Smoothing in your argumentation.

Assignment 3) Laplace Smoothing

(8+2Pts)

In this exercise, we will explore some properties of the Laplace operator and see how smoothing is performed in a more theoretical sense. You may use numpy or other tools of your choice to calculate the results. In the lecture you have seen that the Eigenvalues of the Laplace operator are tightly connected to the Fourier basis functions.

- a) Explain how you could implement a low-pass filter of real valued functions $f : \mathbb{R} \rightarrow \mathbb{R}$ using Fourier Transforms. What is the effect of a low-pass filter?

Consider the polygon

$$\mathcal{P} = \{(0, 1), (2, 2), (1, 0), (-2, -2)\}.$$

Our goal is to smooth this polygon using an explicit low-pass filtering by eliminating high frequency components.

- b) Sample the edges of the polygon such that you have 5 points per edge (i.e., 20 points in total) and calculate the 20×20 Laplace matrix according to `atcg2-07-Mesh_Smoothing.pdf` Slide 25.
- c) Now, we have to represent the polygon using the basis functions of the Laplace Operator. Compute the Eigenvectors of the Laplace matrix

$$L = V\Lambda V^T$$

and project the points of the polygon into the obtained eigen space V .

Hint: Make sure that the eigenvectors are sorted according to the descending eigenvalues.

- d) The result of exercise c) will be two vectors $\hat{x}, \hat{y} \in \mathbb{R}^{20}$ for the x - and y -coordinates, respectively. If we take these vectors and project them back using the inverse transformation, we would obtain the original polygon. To implement the low-pass filter, only use the first three Eigenvectors and the first three components of $\hat{x}$ and $\hat{y}$, respectively. Plot the resulting polygon and discuss the results.
- e) **Bonus:** Implement Taubin Smoothing with 400 iterations, $\lambda = 0.6$ and $\mu = -0.62$. Compare the resulting polygon to the low-pass filtered result.

Good luck!